

YUPING LIN

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EDUCATION

Michigan State University

Ph.D. in Computer Science

09/2023 - 05/2028 (anticipated)

- Advisor: Dr. Jiliang Tang
- Research Area: Safety and Interpretability in Large Language Models, Trustworthy AI

University of Massachusetts, Amherst

M.S. in Electrical and Computer Engineering

01/2022 - 09/2023

- Advisor: Dr. Tongping Liu
- Research Area: Machine Learning Model Compression

University of Electronic Science and Technology of China

B.Eng in Information Security

09/2017 - 07/2021

RESEARCH EXPERIENCE

Crafting Reversible SFT Behaviors in Large Language Models

Ongoing

- We investigate whether Supervised Fine-Tuning (SFT) induced behaviors can be deliberately compressed into sparse, mechanistically necessary subnetworks, termed carriers, that remain controllable at inference time without weight modification.
- We propose Loss-Constrained Dual Descent (LCDD) to construct such carriers by jointly optimizing routing masks and model weights under an explicit utility budget, and SFT-Eraser, a soft prompt optimized via activation matching, to reverse the SFT-induced behavior.
- Our approach yields sparse carriers that preserve target behaviors while enabling strong reversion when triggered, providing direct evidence that the learned carriers are causally necessary for the behaviors.

Understanding Dynamic Retrieval Heads in LLMs

ACL 2026 (Main)

- Existing works identified a special class of attention heads in LLMs responsible for retrieval, termed retrieval heads. However, existing work approaches this from a statistical perspective, overlooking the dynamic nature of these heads as generation steps change.
- Our ongoing investigation has uncovered several key properties of these heads: dynamic, irreplaceable, and predictable. Simultaneously, we extend the exploration of retrieval heads from the constrained Needle-in-a-Haystack test to reasoning tasks.
- We further propose a new in-context retrieval framework and explore its performance in solving Question-Answering tasks.
- Our work, titled *Retrieval Heads are Dynamic*, has been accepted for poster presentation at **ACL 2026 (Main Conference)**.

Memory Management in LLM-based Agents

ACL 2026 (Main, Oral)

- The memory module is a crucial component of LLM-based agents, yet there is no consensus on its optimal design.
- We investigate the general principles of memory management for LLM agents by examining the impact of two core operations—the addition and deletion of memories—on LLM agent performance.
- We examine various memory addition and deletion strategies, and find that with the optimal memory addition strategy, the agent’s execution success rate can increase from 13% to 38%. Additionally, the optimal memory deletion strategy reduces the memory size from 1000 to 250, highlighting the critical importance of effective memory management.
- Our work, titled *Towards Optimal Memory Management: Investigating Experience-Following Behavior of Large Language Model Agents*, has been accepted for poster presentation at **ACL 2026 (Main Conference)**.

Understanding Jailbreak Attacks in LLMs

EMNLP 2024 (Main)

- Although various jailbreak attacks have been developed in the literature, a comprehensive understanding of their underlying mechanisms remains elusive. This gap hinders the advancement of robust jailbreak attacks as well as the development of effective alignment and defense strategies.
- We propose an analytical framework that leverages the hidden representations of LLMs to examine the behavior of jailbreak attacks. Our analysis reveals a clear separation between harmful and benign prompts in the hidden space, with successful jailbreak attacks shifting harmful prompt representations toward the benign direction.

- We demonstrate that steering the representation of jailbreak attacks toward the benign region increases the attack success rate from 26% to 62%, thereby exposing a critical vulnerability in current LLM architectures.
- Our work, titled *Towards Understanding Jailbreak Attacks in LLMs: A Representation Space Analysis*, has been accepted for poster presentation at **EMNLP 2024 (Main Conference)**.

PUBLICATIONS AND PREPRINTS

[*: Equal Contribution]

1. **Yuping Lin**, Jiayuan Ding, Yue Xing, Pengfei He, Jiliang Tang, Subhabrata Mukherjee. *A Simple Plug-in for Improving Eviction-Based KV Cache Compression*. [ICML Workshop of Mechanistic Interpretability. Submitted to NeurIPS 2026]
2. **Yuping Lin**, Pengfei He, Yue Xing, Yingqian Cui, Jiayuan Ding, Subhabrata Mukherjee, Hui Liu, Zhen Xiang. *Crafting Reversible SFT Behaviors in Large Language Models*. [Preprint. Submitted to NeurIPS 2026]
3. **Yuping Lin**, Zitao Li, Yue Xing, Pengfei He, Yingqian Cui, Yaliang Li, Bolin Ding, Jingren Zhou, Jiliang Tang. *Retrieval Heads are Dynamic*. [ACL 2026 (Main)]
4. **Yuping Lin***, Zidi Xiong*, Wenya Xie*, Pengfei He, Jiliang Tang, Himabindu Lakkaraju, Zhen Xiang. *How Memory Management Impacts LLM Agents: An Empirical Study of Experience-Following Behavior*. [ACL 2026 (Main, Oral)]
5. Tai-Quan Peng, Kaiqi Yang, Sanguk Lee, Hang Li, Yucheng Chu, **Yuping Lin**, Hui Liu. *Beyond partisan leaning: A comparative analysis of political bias in large language models*. [Journal of Information Technology & Politics, 2026]
6. Shenglai Zeng, Jiankun Zhang, Bingheng Li, **Yuping Lin**, Tianqi Zheng, Dante Everaert, Hanqing Lu, Hui Liu, Yue Xing, Monica Xiao Cheng, Jiliang Tang. *Towards Knowledge Checking in Retrieval-augmented Generation: A Representation Perspective*. [NAACL 2025]
7. Yingqian Cui, Jie Ren, **Yuping Lin**, Han Xu, Pengfei He, Yue Xing, Lingjuan Lyu, Wenqi Fan, Hui Liu, Jiliang Tang. *FT-Shield: A Watermark Against Unauthorized Fine-tuning in Text-to-Image Diffusion Models*. [SIGKDD Explorations Newsletter, 2025]
8. Pengfei He, **Yuping Lin**, Shen Dong, Han Xu, Yue Xing, Hui Liu. *Red-Teaming LLM Multi-Agent Systems via Communication Attacks*. [ACL 2025]
9. **Yuping Lin***, Pengfei He*, Han Xu, Yue Xing, Makoto Yamada, Hui Liu, Jiliang Tang. *Towards Understanding Jailbreak Attacks in LLMs: A Representation Space Analysis*. [EMNLP 2024]

INTERNSHIPS

Applied Science Intern, Amazon Development Center U.S., Inc.

Summer 2026

- Worked on research of Interpretability in LLM Agents.
- Worked on the investigation of a novel attention head taxonomy and the development of a visual auditing tool of LLM Agent trajectories.

Intern, Hippocratic AI Inc.

Spring 2026

- Worked on research of KV Cache Compression in Large Language Model inference.

Research Intern, Alibaba Group US Inc.

Summer 2025

- Worked on the research of Long-Context processing in LLM Agents.
- Developed an open-source MCP toolkit for LLM Agent Long-Context handling.

SERVICES

- Emergency Reviewer for ARR January 2026
- Reviewer for ARR January 2026
- Reviewer for ARR October 2025
- Reviewer for Computational Linguistics, 2025

TEACHING

Teaching Assistant

Spring 2025

CSE 335 Object-oriented Software Development

Teaching Assistant

Fall 2024

CSE 482 Big Data Analysis

AWARDS

- Travel grant, 2024, Michigan State University
- Outstanding Student Scholarship in the academic year of 2018, 2019, UESTC
- Third prize in the 16th UESTC ACM Programming Competition, 2018
- CCF National Olympiad in Informatics 2015 (2nd), 2016 (2nd)
- Second prize in Computer Production Activities for Primary and Secondary schools in Shandong Province, 2016

PROFESSIONAL SKILLS

- **Programming Languages:** Python, C, C++, Pascal, JavaScript, Java, HTML/CSS, MATLAB
- **Softwares and Systems:** PyTorch, HuggingFace, Numpy, Pandas, Scikit-Learn, Git, Docker, Linux, Vue, SpringBoot
- **AI skills:** Large Language Models (LLaMA, DeepSeek, Qwen, Gemma, Mixtral, Vicuna, GPT), Computer Vision (Diffusion models), Data Science, Data Visualization